

Degradation of an organic pollutant in water using LED light and a photocatalyst

BS 192, Experiment: 5

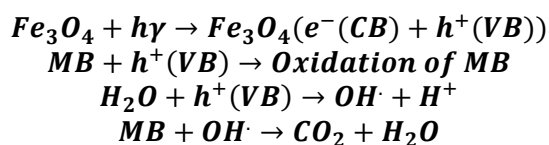
We will use light to degrade organic dye molecules, which are commonly used in dye industries. They are often present in the industrial wastewater. We will use regular LED torches and a photocatalyst to remove/degrade them. We will study the kinetics of the reaction. We will also understand the usefulness of a photocatalyst.

Materials:

Methylene blue and Iron Oxide Nanoparticles (Fe_3O_4)

Basic principle:

The photocatalyst absorbs light and produces holes and electrons which helps in degrading the dyes. The nanoparticle may also adsorb some dyes due to electrostatic interactions.



Procedure:

1. Prepare three vials with the following details:

- (i) Vial 1: 5 ml methylene blue (stock solution) + 5 mg of Fe_3O_4 nanoparticles
- (ii) Vial 2: 5 ml methylene blue (stock solution) + 5 mg of Fe_3O_4 nanoparticles + 1 magnetic bead
- (iii) Vial 3: 5 ml methylene blue

2. Wrap vial 1 with Al foil and keep aside

3. Put vial 2 and 3 under the LED light on top of the magnetic stirring bar

4. In every 10 minutes:

(i) Put vial 1 near the magnetic post/bar to remove the nanoparticles and then take 2 ml of the sample and measure absorbance spectrum. Pour this 2 ml back in the vial (after measurement) and keep it in dark.

(ii) Take vial 2 out and put near the magnetic post/bar to remove the nanoparticles and then take 2 ml of the sample and measure absorbance spectrum. Pour this 2 ml back in the vial (after measurement) and put the vial back under LED at the earliest.

(iii) Take 2 ml from vial 3 and measure absorbance spectrum. Pour this 2 ml back in the vial (after measurement) and put the vial back under LED at the earliest.

Through this measurement, you will have absorbance from 3 vials every 10 minutes.

5. Determination of concentration from absorbance:

According to Beer-Lambert's law:

$$\text{Absorbance} = \epsilon \cdot c \cdot l$$

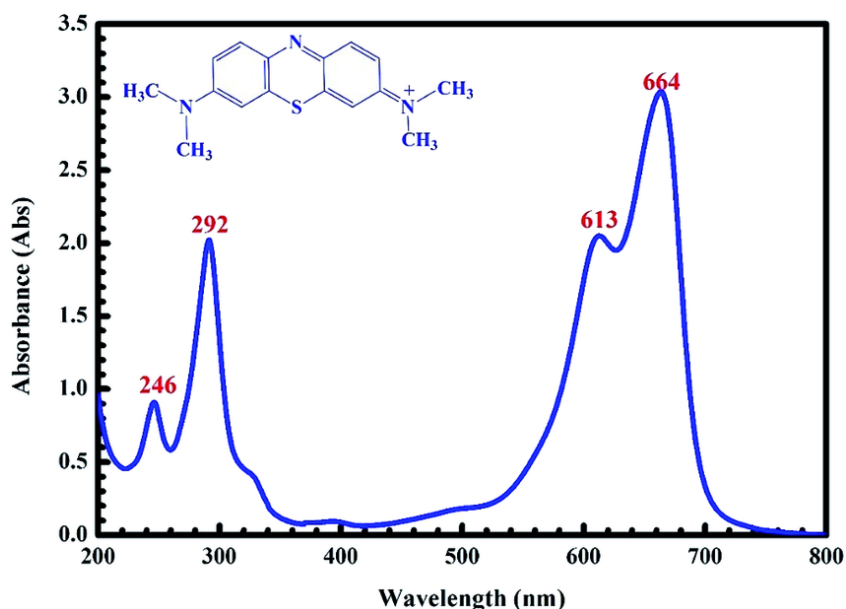
Where ϵ = molar extinction coefficient, a property of the molecule being studied. C is the concentration and l is the path length. For a given molecule and a given experimental set up, ϵ and l are constant. Thus absorbance will be proportional to concentration.

5.1. Calibration with known concentrations:

From a given MB solution of 30 ppm concentration prepare the following 5 test tubes (TT):

- (i) TT 1: 2 ml MB
- (ii) TT 2: 1 ml MB + 1 ml water
- (iii) TT3: 1 ml MB + 2 ml water
- (iv) TT4: 1 ml MB + 3 ml water
- (v) TT5: 1 ml MB + 4 mL water

Take ~ 2ml from each of the test tube and measure their absorption spectra. Plot absorbance at the peak (~664 nm, typical spectrum of MB is given below) with MB concentrations. From the slope of this graph, calculate ϵ . Note that $l=1$ cm.



5.1. calculate concentrations of MB in vial1, 2, and 3 at different time (refer to point 4 in procedure) using the values of ϵ and l .

6. Plot $\ln(C_0/C)$ vs t for vials 1,2, and 3. Calculate the slopes from each graph. C_0 is the concentration of MB at $t=0$ time. Comment on the kinetics of the reactions.

7. Calculate the amount of MB degraded after 20 minutes of light exposure in presence of the catalyst and compare this value with the situation where MB was exposed to light in absence of the catalyst.

Your report should include the following data:

- All absorption spectra measured from vials 1,2, and 3. One plot for each of the vial, showing the decrease of absorbance with respect to time.
- Absorption spectra measured from test tubes 1-5. Plot of concentration vs absorbance. Calculation of ϵ **value of MB**.
- $\log(C_0/C)$ vs t for vials 1,2,3. These three plots should be reported in 1 single figure.